🌡️ Temperature Monitoring System using Arduino

📌 Objective

To measure temperature using an LM35 sensor and display real-time values on a 16x2 LCD while logging readings to the Serial Monitor.

🧰 Components Used

- Arduino Uno
- LM35 Temperature Sensor
- 16x2 I2C LCD
- Breadboard
- Connecting Wires

🔌 Circuit Connections

LM35 Sensor

- VCC → 5V
- GND → GND
- OUT → A0

I2C LCD

- VCC → 5V
- GND → GND
- SDA → A4
- SCL → A5

⚙️ Working Principle

The LM35 sensor produces an analog voltage proportional to temperature (10 mV per °C). The Arduino reads this voltage using analog pin A0, converts it into temperature in Celsius, and

displays it on the LCD. The same data is also printed on the Serial Monitor for logging and debugging.

💻 Arduino Code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// LCD setup
LiquidCrystal_I2C lcd(0x27, 16, 2);

int lm35Pin = A0;

// Function to get stable temperature
float getTemperature() {
  int sum = 0;

  for (int i = 0; i < 10; i++) {
    sum += analogRead(lm35Pin);
    delay(10);
  }

  float avg = sum / 10.0;
  float voltage = avg * (5.0 / 1023.0);
  float temp = voltage * 100.0;

  return temp;
}
void setup() {
  Serial.begin(9600);

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0, 0);
  lcd.print("Temp Monitor");
  delay(2000);
  lcd.clear();
}

void loop() {
  float temperature = getTemperature();
```

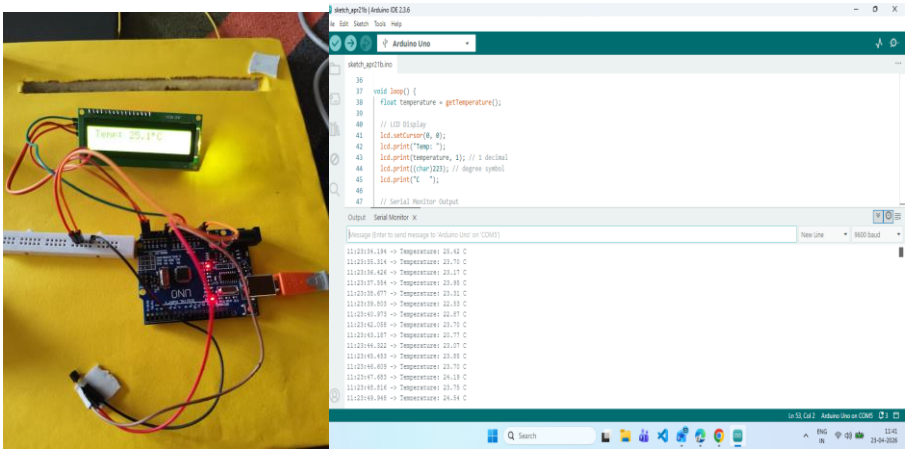
```
  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print(temperature, 1);
  lcd.print((char)223);
  lcd.print("C   ");

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" C");

  delay(1000);
}
```

Output

- LCD displays real-time temperature (°C)
- Serial Monitor logs temperature every second

Screenshots (Attached)

- Circuit setup image
- Arduino IDE code
- LCD displaying temperature
- Serial Monitor output



Demonstration

Due to time constraints, a video demonstration could not be recorded. However, the project was successfully tested, and screenshots are provided as proof of working.

Conclusion

This project demonstrates sensor interfacing, analog-to-digital conversion, and real-time data display using Arduino. It is a simple and effective system for temperature monitoring applications.